COS20019-Cloud Computing Architecture



Assignment 3: A Serverless and Event-Driven Architecture for a Globally Scalable Photo Album

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Abstract - The Photo Album workload is growing rapidly, with demand expected to double every six months. This report proposes a serverless, event-driven AWS design that removes large media transfers from the web application path and decouples each transformation. Route 53, CloudFront, AWS WAF and Shield Standard provide a global edge. Cognito authenticates users, an API Gateway HTTP API invokes Lambda business logic, and DynamoDB stores media metadata. A short-lived presigned URL lets the client upload directly to Amazon S3, optionally through S3 Transfer Acceleration. S3 ObjectCreated events are routed through EventBridge to separate SQS queues for image, video and AI jobs. Lambda handles image work, a dedicated Lambda submitter creates MediaConvert jobs, and an optional Rekognition consumer detects image labels. Each path scales independently. The design uses retries, dead-letter queues, idempotent updates, versioning, PITR, least-privilege IAM and CloudWatch alarms. Under the stated baseline, the estimated cost is USD 1,032 per month and reaches about USD 56,313 at 64x demand, while unit cost falls because of pricing tiers. The selected design scores 4.65/5 in a weighted comparison against EC2 and containers.

Index Terms - *AWS, serverless, event-driven architecture, Amazon S3, EventBridge, SQS, Lambda, DynamoDB, Rekognition, CloudFront.*

I. INTRODUCTION

The current Photo Album has been successful, but its six t3.micro instances regularly exceed 80% utilization, and overseas response time is poor. The new architecture must adopt managed services, store media in S3, support uncertain growth, reduce database cost, automatically create media variants, and remain extensible for video and AI-based photo tagging. These requirements favor a design that separates the synchronous user request from long-running media work. The proposal follows the AWS Well-Architected pillars of security, reliability, performance efficiency and cost optimization [1], [2].

The report is written for both technical and business stakeholders. It first states assumptions, then traces each business requirement to a service, explains the architecture and service choices, documents the upload and processing interactions, and finally compares alternatives with quantitative estimates.

A. Assumptions and Quality Targets

ID	Assumption	Value
A1	Primary Region	Asia Pacific (Sydney), ap-southeast-2. Global edge services remain outside the Region boundary.
A2	Baseline demand	50,000 monthly active users (MAU), 10 million API calls and 100,000 photo uploads per month.
A3	Media profile	Average original is 5 MB; two image variants are created. Steady-state S3 storage is 1 TB.
A4	Global delivery	5 TB/month CloudFront transfer, 50 million HTTPS media requests and a 90% cache-hit target.
A5	Performance	API p95 < 400 ms (excluding file transfer); image processing p95 < 30 s; queue age < 5 min.
A6	Recovery	Multi-AZ service failures are handled automatically. Regional DR is a later warm-standby phase with target RTO 4 h/RPO 15 min.
A7	Data model	Media metadata and optional AI tags are key-value/document shaped; cross-row relational transactions and ad-hoc joins are not required.
A8	Estimate basis	Public AWS list prices on 28 July 2026, USD, before tax and support. Promotional credits are ignored.

B. Requirements Traceability

Req.	Need	Design response
1	Managed services and S3	API Gateway, Lambda, DynamoDB, S3, EventBridge, SQS, CloudFront and managed security/observability.
2-3	Rapid growth; EC2 >80%	Remove EC2 from the online path. Serverless capacity scales per request; SQS controls worker concurrency.
4	Serverless/event-driven	S3 ObjectCreated -> EventBridge -> SQS -> specialised consumer.
5	Slow/costly relational DB	DynamoDB on-demand for simple metadata/status with no idle database instance.
6	Slow global response	Route 53 + CloudFront CDN; cache media near users and shield origins.
7	Future video	Dedicated video queue and MediaConvert consumer can be enabled without changing upload/API services.
8a	Automatic variants	S3 ObjectCreated event automatically starts the processing pipeline.
8b	Extensible processing	EventBridge rules and independent queues add consumers without changing producers.
8c	Best execution platform	Lambda for short image work, MediaConvert for video and Rekognition for optional photo labels; ECS/Batch remains an option for custom long-running jobs.
8d	Decoupled/overload safe	SQS buffers bursts, supports visibility timeouts/retries/DLQs, and lets workers scale from queue depth.

II. PROPOSED ARCHITECTURE

Figure 1 combines two deliberately different flows. The synchronous path authenticates a user, creates an upload session and serves metadata. The asynchronous path starts only after S3 confirms object creation. Because the uploaded bytes do not pass through API Gateway or Lambda, a large or slow upload cannot occupy application compute. CloudFront also keeps most media reads away from S3.

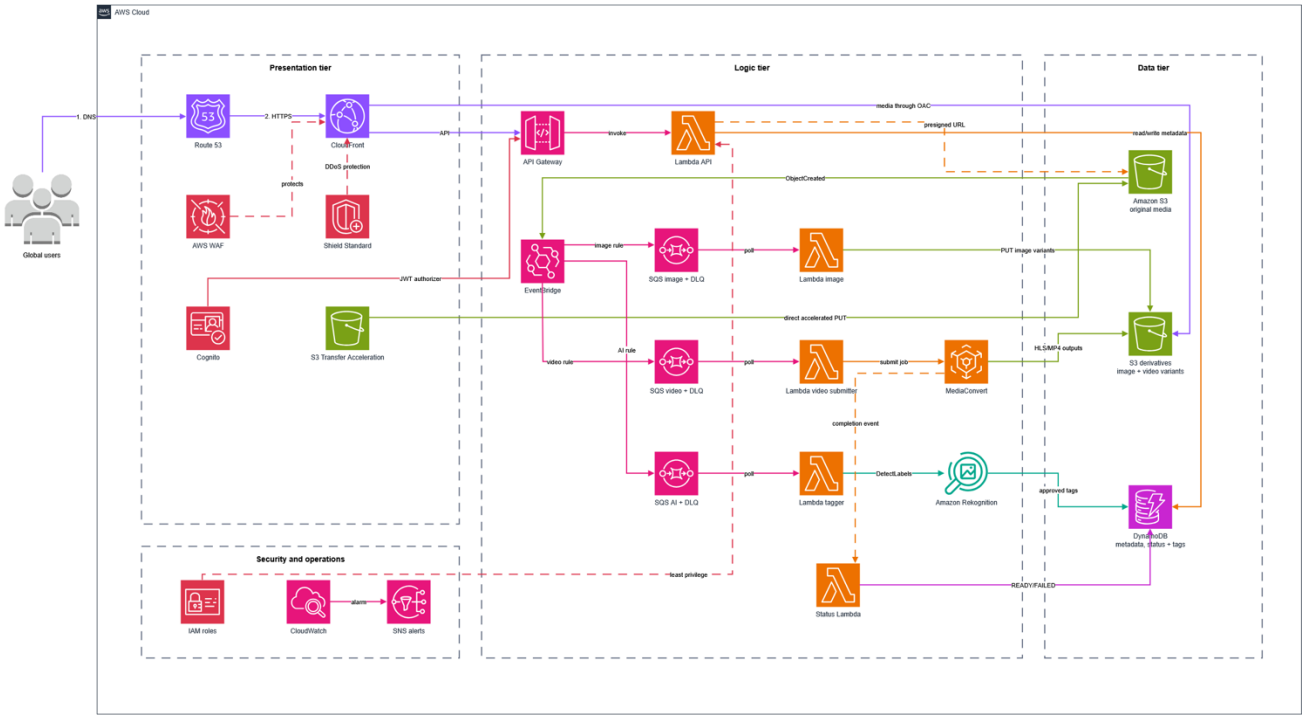


Fig. 1. Proposed serverless/event-driven architecture and deployment scope.

Deployment scope is explicit. Route 53, CloudFront, CloudFront-associated WAF and Shield Standard are global. Cognito, API Gateway, Lambda, DynamoDB, S3, EventBridge, SQS, MediaConvert, Rekognition, IAM, CloudWatch and SNS are regional managed services in ap-southeast-2. No service is drawn inside a VPC because this version has no customer-managed network resources. Keeping Lambda outside a VPC avoids NAT Gateway cost and unnecessary network attachment. If a later custom codec requires ECS or EC2, only that worker should be placed in private subnets with an S3 VPC endpoint.

Decoupling occurs at three boundaries. First, the client uploads directly to S3 using a time-limited presigned URL [5]. Second, EventBridge performs content-based routing rather than embedding every target in the uploader [3]. Third, each SQS queue stores work until its consumer has capacity. Image, video and AI consumers therefore fail, scale and deploy independently. The AI queue can remain disabled until its privacy policy and budget are approved. The event contains identifiers, not media bytes; workers read the object from S3.

III. SERVICE DESCRIPTIONS AND SELECTION RATIONALE

A. Edge, Access and Application Services

Service	Function	Why it is used
Route 53	Public DNS alias to CloudFront.	Highly available managed DNS. Alias queries to CloudFront are not charged.
CloudFront	Caches the web UI and media; forwards /api/* to API Gateway.	Reduces overseas latency, origin GETs and application load. OAC keeps the S3 origin private [6].
AWS WAF + Shield Standard	Blocks common exploits, applies rate limits and provides baseline DDoS protection.	Security is enforced before requests reach the regional API or S3 origin.
Cognito User Pool	Registration, sign-in and JWT issuance.	Removes password storage from the application and integrates with API Gateway authorisation.
API Gateway HTTP API	TLS API endpoint, route matching, throttling and JWT validation.	Lower-cost serverless API surface; upload bodies are not sent through it.
Lambda API	Validates metadata, creates media IDs/presigned URLs and queries status.	Scales per request and has no idle instance. Reserved concurrency protects downstream services [7].
DynamoDB on-demand	Stores owner, object keys, status, checksum, timestamps and approved AI tags.	The model is simple and access is key-based. On-demand mode scales automatically and charges per request [8].
Amazon S3	Private buckets/prefixes for originals, derivatives and quarantine.	Durable object storage with versioning, lifecycle rules and EventBridge integration.
S3 Transfer Acceleration	Optional accelerated upload endpoint using AWS edge locations.	Improves long-distance uploads. The client can fall back to the normal regional endpoint if acceleration is not faster.

B. Event Processing, Security and Operations Services

Service	Function	Why it is used
EventBridge	Receives durable S3 ObjectCreated events and matches bucket, prefix, suffix and metadata.	A new consumer is added as a rule/target; the uploader remains unchanged [3].
SQS Standard + DLQ	Buffers one queue per workload, retries failures and isolates poison messages.	Pull-based back-pressure. Lambda receives batches and failed records can be retried or moved to a DLQ [4].
Lambda image worker	Reads an original, verifies the version/checksum and creates thumbnail/mobile variants.	Short, parallel, bursty work fits Lambda. A concurrency cap prevents a burst overwhelming S3 or libraries.

Service	Function	Why it is used
Lambda video + MediaConvert	The Lambda consumer validates a video message and submits an SD/HD or HLS/MP4 job.	SQS cannot submit a job by itself. A small submitter preserves queue isolation, while the managed transcoder avoids maintaining video worker fleets.
Lambda tagger + Rekognition	The optional consumer calls DetectLabels and stores labels above an approved confidence threshold.	A managed vision service provides future photo AI without training infrastructure. A separate queue prevents tagging latency or failures affecting required resize work [19].
Status Lambda	Consumes completion events and conditionally writes READY/FAILED.	Centralizes an idempotent state transition and prevents duplicate events overwriting a newer state.
IAM	Defines least-privilege roles and policies for each application and processing service.	Limits the actions and S3 prefixes available to each component. S3 Block Public Access and CloudFront OAC prevent direct public reads.
CloudWatch + SNS	Logs, metrics, dashboards, queue-age/DLQ alarms and operator notification.	Provides evidence for performance, reliability and cost tuning; alarms make failures actionable.

IV. COLLABORATION / SEQUENCE DIAGRAMS

A. User Authentication

The application uses the Cognito User Pool secure remote password (SRP) flow. The password is entered in the client and is not stored by the Photo Album API. After the SRP challenge is completed, Cognito returns short-lived ID and access tokens plus a refresh token. API Gateway later validates the JWT issuer, audience and expiry before Lambda is invoked.

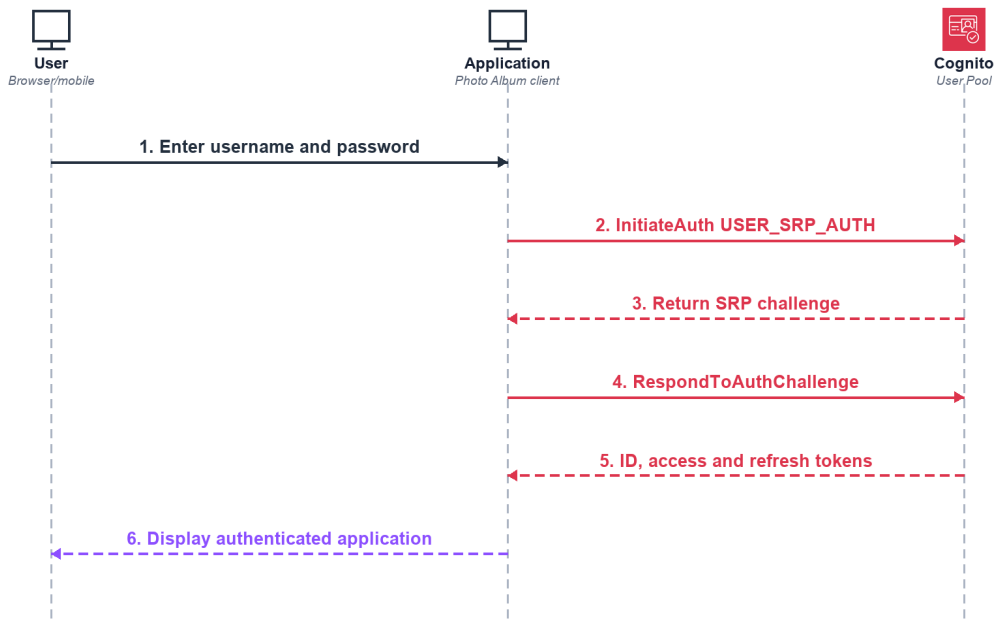


Fig. 2. User authentication and token issuance through Amazon Cognito.

B. Authenticated Media Upload

The browser first obtains Cognito tokens, then requests an upload session using only metadata. Lambda writes UPLOADING and returns a short-lived presigned PUT URL. The browser uploads directly to S3 and includes a checksum. An S3 ObjectCreated event is the authoritative asynchronous hand-off. A unique, non-guessable key such as users/{sub}/{mediaId}/original prevents accidental overwrite; the bucket policy also limits presigned signature age and enforces TLS.

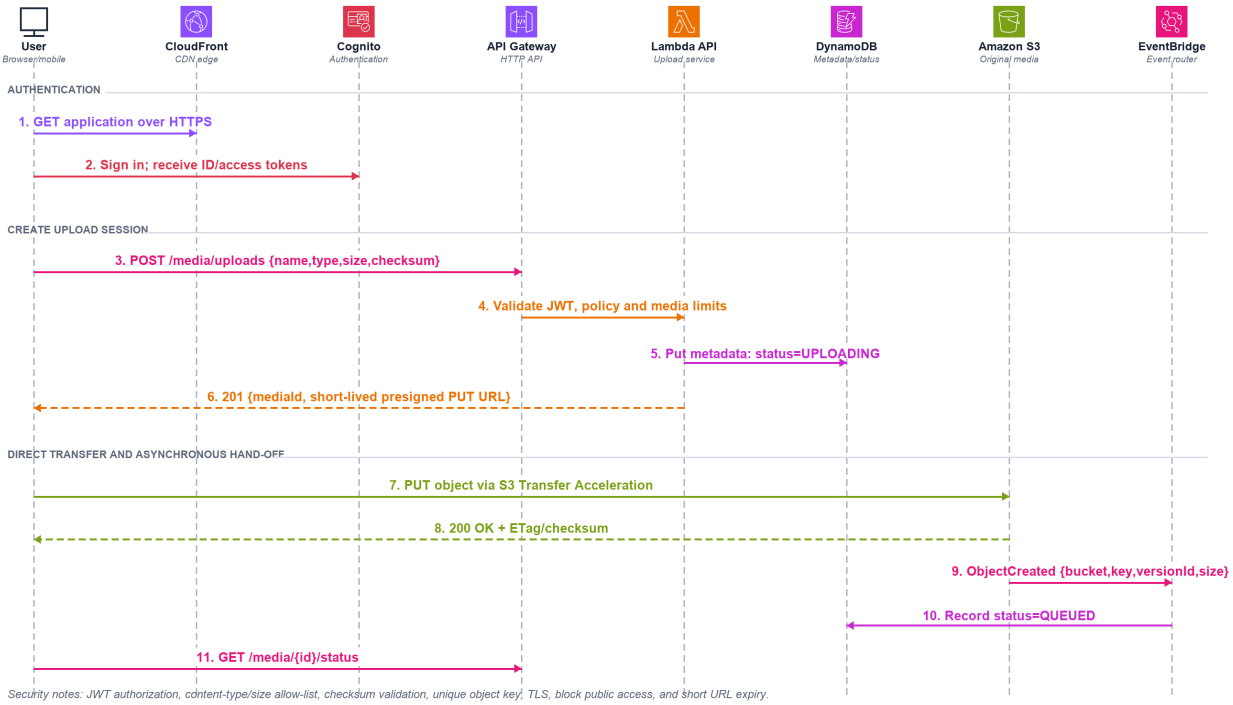


Fig. 3. Ordered interactions for the authenticated media upload use case.

The 201 responses do not mean processing is complete. The UI polls GET /media/{id}/status or may later use a WebSocket/SNS-to-notification extension. Returning a media ID immediately gives the user a responsive experience even when image/video work takes minutes.

C. Image, Video and AI Processing

S3 notifications can be duplicated and are not guaranteed to arrive in order [3]. The object version ID and media ID are therefore used as the idempotency key. EventBridge selects a queue; a consumer polls the queue, processes the media, writes immutable derivative keys and performs a conditional status update. For partial batch failure, only failed SQS records are returned. After maxReceiveCount = 5, the message moves to a DLQ and an alarm opens an operational task.

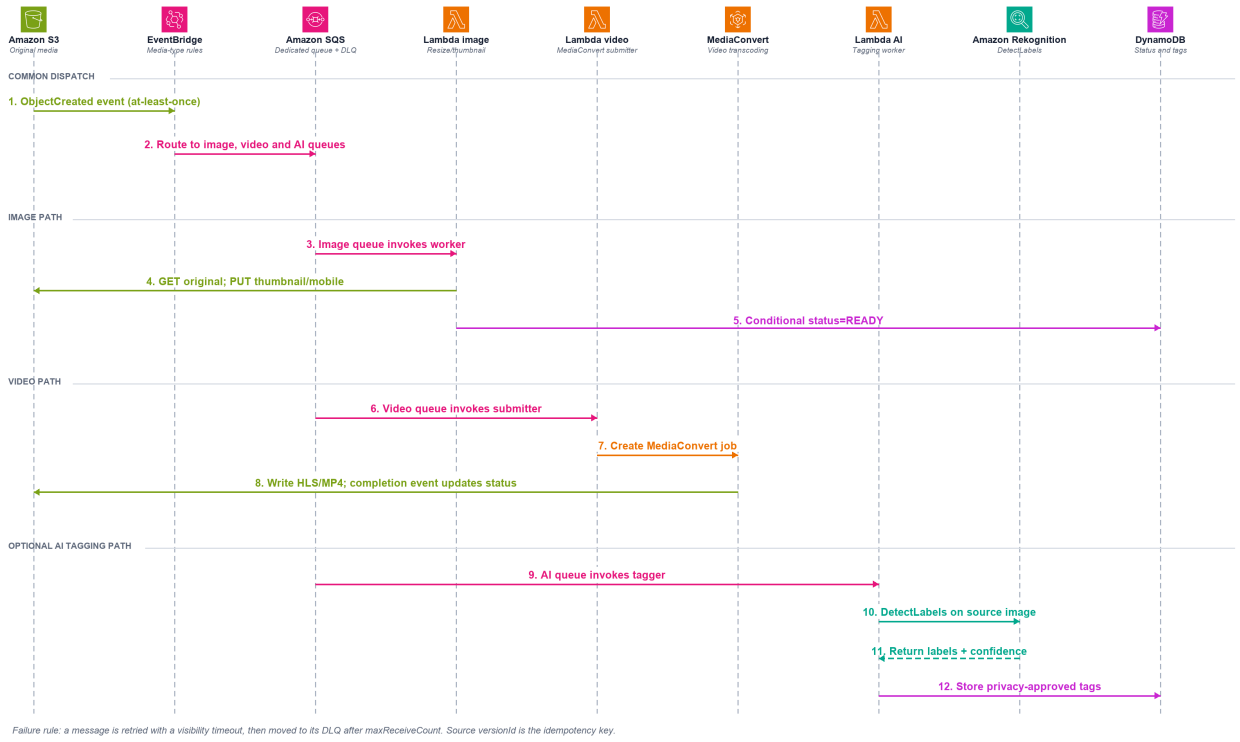


Fig. 4. Event-driven image, video and optional AI processing interactions.

Image transformation is the initial required path. The video consumer only submits a MediaConvert job; MediaConvert writes the output and emits a completion event for the Status Lambda. Rekognition is optional and only receives image jobs. Labels are stored after a confidence threshold and privacy policy check. Each queue can use a different timeout, batch size, retry policy, retention period and concurrency limit.

D. Browse and Retrieve Media

Browsing is split into metadata and media delivery. GetMedia queries DynamoDB for items owned by the authenticated user and returns only READY records with CloudFront URLs. When the browser requests a variant, CloudFront checks the edge cache. A hit is returned immediately; on a miss, CloudFront obtains the private object from S3 through Origin Access Control (OAC), caches it, and returns it to the user.

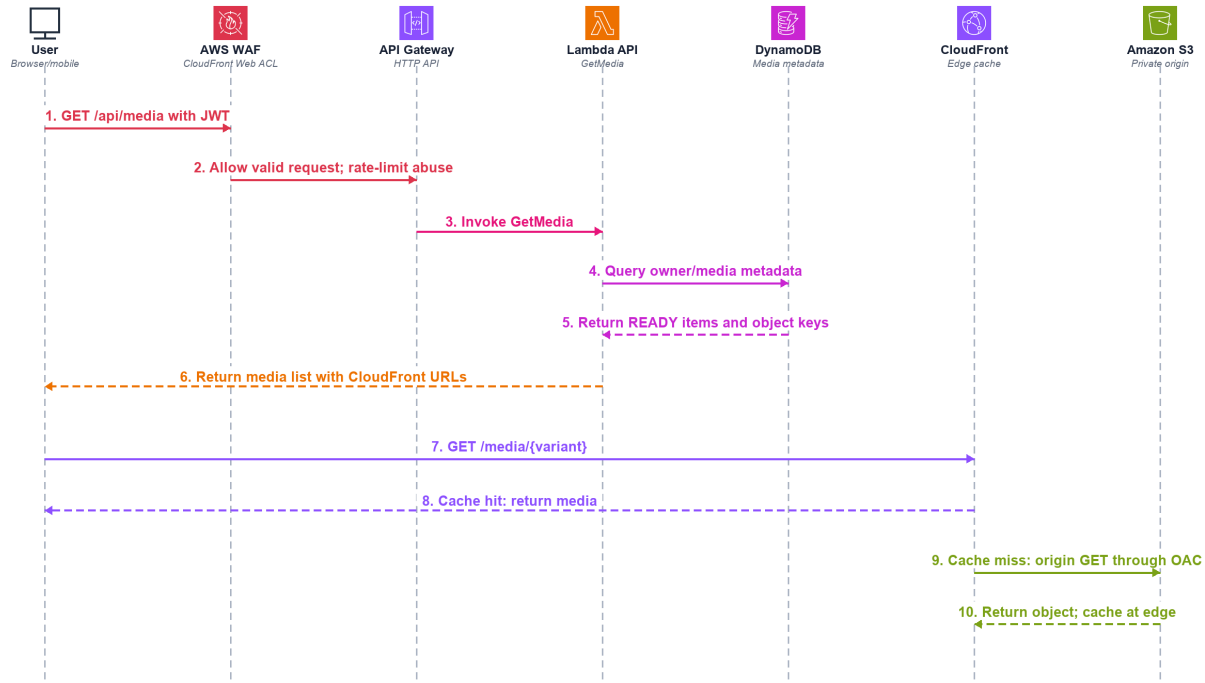


Fig. 5. Metadata retrieval and CloudFront cache hit/miss behavior.

The WAF is associated with CloudFront rather than deployed as an application server. It applies managed rules and rate limits before requests reach either the regional API origin or the S3 media origin.

V. DESIGN RATIONALE

A. Performance, Scalability and Extensibility

Demand doubling every six months produces a 16x workload after two years and a 64x workload after three. CloudFront moves cacheable reads to edge locations, while presigned S3 upload removes multi-megabyte bodies from API Gateway/Lambda. DynamoDB on-demand absorbs unpredictable key-based traffic without a capacity plan. Lambda scales concurrent execution environments automatically, subject to account quotas [7].

$$\text{Growth multiplier:} \quad G(t) = 2^{t/6} \; ; \; G(24) = 16, \qquad G(36) = 64.$$

Back-pressure is more important than unlimited scaling. A reserved concurrency limit on the image worker sets the maximum pressure on S3 and the transformation library. For example, a burst of 10,000 images, 100 concurrent workers and 2 s average processing time drains in about 10,000 / (100/2) = 200 s. A video surge stays in the video queue and does not consume image concurrency. Rekognition requests are also isolated behind the AI queue, so an AI quota or service error cannot delay thumbnails. A future processor can be added with one EventBridge rule, one queue and one consumer, which is the main extensibility benefit.

B. Reliability

The selected regional services are managed across multiple Availability Zones; there is no single EC2 instance, database host or worker queue to repair. S3 Versioning protects against overwrite/delete, and lifecycle policies retain originals separately from disposable derivatives. DynamoDB PITR provides managed continuous backups with up to 35 days of per-second recovery points [9]. SQS visibility timeouts, retries, partial batch responses and DLQs handle transient and permanent failures [4].

Reliability depends on idempotency. Each derivative key includes the source version and transformation profile; a conditional DynamoDB update accepts only the expected state/version. If a worker repeats a message, it either produces the same object or detects that the output already exists. CloudWatch alarms monitor oldest-message age, DLQ depth, Lambda errors/throttles, API 5xx, DynamoDB throttling, MediaConvert failures and Rekognition throttling.

The baseline is intentionally single-Region to control cost and complexity. It tolerates AZ failures but does not guarantee the target during a full Sydney Region outage. If the client confirms a 4 h RTO and 15 min RPO, Phase 2 should add S3 Cross-Region Replication, a DynamoDB Global Table and an API/Lambda stack in a second Region, then use Route 53 or CloudFront origin failover. This limitation is explicit rather than hidden.

C. Security

Cognito issues short-lived JWTs; API Gateway validates issuer, audience and scopes. WAF managed rules, a rate-based rule and Shield Standard reduce edge attacks. The S3 buckets block public access, use Bucket Owner Enforced ownership and allow reads only through CloudFront OAC [6]. Presigned uploads expire after about five minutes, permit one object key and require an allowed content type, maximum size and checksum [5]. Files enter a quarantine prefix and are not served until the status becomes READY.

Every Lambda, MediaConvert and Rekognition role receives only the actions and S3 prefixes it needs. Data in transit uses TLS, while bucket policies and service roles prevent unauthorized object access. CloudTrail records control-plane actions, and CloudWatch log retention is set explicitly. AI tagging is opt-in at deployment: the tagger reads only the source-image prefix, uses an approved confidence threshold, avoids biometric identification, and supports removing tags when a user deletes an album.

D. Detailed Monthly Cost Estimate

Table IV separates fixed from variable charges. The estimate uses public list prices for Sydney or the global service price zone [10]-[19]. CloudFront transfer and Cognito MAU dominate; Lambda, EventBridge and SQS are comparatively small. This result supports a serverless application path but also shows that global media delivery and identity, not compute, should receive the strongest cost controls.

Service	Usage assumption	Fixed USD	Variable USD
Route 53	1 hosted zone; alias to CloudFront	0.50	0.00
AWS WAF	1 Web ACL + 5 rules; 60 M requests	10.00	36.00
CloudWatch metrics	10 custom metrics	3.00	0.00
CloudFront transfer	5,120 GB at $0.114/GB	0.00	583.68
CloudFront HTTPS	50 M requests at $1.25/M	0.00	62.50
Cognito Lite	50 k MAU; 10 k free; 40 k billed	0.00	220.00
API Gateway HTTP API	10 M requests at $1.29/M	0.00	12.90
Lambda API	10 M req; 512 MB x 150 ms (Arm)	0.00	12.00
Lambda image	100 k req; 1 GB x 2 s (Arm)	0.00	2.69
S3 storage	1,024 GB Standard	0.00	25.60
S3 requests	300 k PUT and 5 M origin GET	0.00	3.85
S3 Transfer Accel.	500 GB at conservative $0.08/GB	0.00	40.00
DynamoDB + PITR	8 M RRU, 2 M WRU, 10 GB PITR	0.00	4.84
EventBridge + SQS	100 k events; about 300 k SQS actions	0.00	0.22
CloudWatch logs	20 GB ingest + 20 GB-month storage	0.00	14.06
SNS	Low-volume operator alarms	0.00	0.10
TOTAL	Estimated baseline per month	13.50	1,018.44

Baseline total = $13.50 fixed + $1,018.44 variable = $1,031.94/month. Optional extensions are excluded from the baseline: 100,000 Rekognition image-label calls add approximately $100/month at an illustrative $0.001/image, while 10,000 video output minutes with one SD and one HD AVC rendition add about $255/month before extra storage and delivery [18], [19].

VI. ALTERNATIVE SOLUTIONS AND TRADE-OFFS

A. Compute Platform

Three compute models were scored from 1 (poor) to 5 (strong). Weights reflect the scenario: performance/scalability 25%, reliability 20%, security 15%, operational simplicity 15%, cost 20% and extensibility 5%. The weighted result is not a universal truth; it makes the design priorities visible.

Option	Perf. 25%	Reliab. 20%	Security 15%	Ops 15%	Cost 20%	Extend. 5%	Weighted /5
EC2 Auto Scaling	3	3	3	2	2	3	2.65
ECS/EKS containers	4	4	4	3	3	4	3.65
Selected serverless	5	4	4	5	5	5	4.65

EC2 can run any library and may be economical at steady utilisation, but requires patching, AMIs, load balancing and scaling policies. Containers improve portability and suit a custom long-running codec, yet EKS adds a control plane and cluster operations. Serverless is selected because the workload is bursty and the company explicitly wants managed/event-driven services. It also lets each media type use its best platform. Containers remain a valid worker target behind SQS if Lambda or MediaConvert cannot meet a future codec requirement.

B. Database, Caching, Messaging and Tiering

Decision	Alternative	Selected	Reason
Database	RDS/Aurora SQL	DynamoDB on-demand	SQL is preferable for joins and complex transactions. The stated table is simple, so DynamoDB avoids instance cost and scales with demand.

Decision	Alternative	Selected	Reason
Caching	API Gateway cache or ElastiCache	CloudFront; no API cache initially	Media dominates volume and is globally cacheable. API Gateway cache has fixed hourly cost; ElastiCache requires a VPC. Add only after measurements show a hot metadata key.
Event trigger	Direct S3 -> Lambda	S3 -> EventBridge -> SQS	Direct trigger is simpler but tightly couples one consumer and has weaker workload isolation. The selected pattern separates routing from buffering.
Push/pull	SNS push	SQS pull by workers	SNS gives low-latency fan-out, but a slow consumer can be retried aggressively. SQS lets consumers control rate and exposes backlog age.
Photo AI	Custom model on SageMaker/ECS	Rekognition DetectLabels	A custom model gives more domain control but needs labelled data, training and operations. Rekognition is appropriate for general labels; a custom model remains an option if accuracy tests fail.
Tiers	Traditional 3-tier EC2	Logical edge/API/data tiers on managed services	Separation is retained without always-on web, application and database servers.
Global design	Active-active multi-Region from day one	Global edge + single Sydney Region	Active-active reduces regional outage but doubles regional resources and data consistency complexity. Add when RTO/RPO justifies it.

VII. QUANTITATIVE EVALUATION

A. Capacity and Concurrency

The existing six t3.micro instances at at least 80% use provide 4.8 instance-units of work. To reach the midpoint of the required 50-60% range, an EC2 design needs at least nine instances now. If capacity scaled linearly with 64x demand, it would need approximately 559 t3.micro instances. At the Sydney on-demand price of $0.0132 per hour, compute alone would be about $86.72/month now and $5,387/month at 64x, excluding load balancers, database, storage, networking and administration.

$$N_{\text{now}} = \left\lceil \frac{6 \times 0.80}{0.55} \right\rceil = 9; \qquad N_{64\times} = \left\lceil \frac{6 \times 0.80 \times 64}{0.55} \right\rceil = 559.$$

For the serverless baseline, 10 million API requests/month is 3.86 requests/s on average. At 150 ms, average Lambda API concurrency is 0.58; at 64x it is about 37, or about 370 under a 10x burst. Image concurrency is about 0.08 now and 4.9 at 64x average, or about 49 under a 10x burst. The combined illustrative burst remains below the default 1,000 regional concurrency quota, but quota and reserved-concurrency settings must be load-tested.

B. Three-Year Cost Forecast

Horizon	Load	MAU	CDN TB	API M	Monthly USD	USD/MAU
Now	1x	50,000	5	10	1,032	0.0206
24 months	16x	800,000	80	160	15,234	0.0190
36 months	64x	3,200,000	320	640	56,313	0.0176

The forecast applies current CloudFront, Cognito, API Gateway and S3 pricing tiers and scales the remaining variable assumptions. Optional Rekognition and MediaConvert usage are excluded because the brief does not specify adoption volume; their unit add-ons are shown below Table IV. It is a planning model, not a quotation. Cost grows to 54.5x when demand grows 64x, so unit cost declines, but the absolute budget is material. At

higher scale the team should compare CloudFront flat-rate plans/private pricing, tune cache keys, use S3 lifecycle tiers, batch SQS operations, and revisit Cognito or identity-provider economics. AWS Budgets should alarm at 80%, 100% and forecasted 120% of budget.

C. Verification Plan and Acceptance Measures

Area	Test	Acceptance measure
Performance	Load test 10x peak API and 10,000-object upload burst.	API p95 <400 ms; image queue drains <5 min; CloudFront hit ratio >=90%.
Reliability	Inject worker failures, duplicates, out-of-order events and a poison file.	No lost media; exactly one final state; poison item reaches DLQ after 5 receives; replay succeeds.
Security	Attempt expired/wrong-key presigned PUT, public S3 GET, oversized type and invalid JWT.	All denied and recorded; published media accessible only through authorised CloudFront path.
Cost	Compare Cost Explorer tags and CUR with the model for two weeks.	Forecast error <20%; alert before any service exceeds its monthly guardrail.
Recovery	Restore DynamoDB PITR and an S3 prior version; run IaC rebuild.	Evidence of data recovery; baseline regional rebuild <4 h once Phase 2 DR is funded.

VIII. RISKS, IMPLEMENTATION AND LIMITATIONS

Risk	Impact	Mitigation
Duplicate/out-of-order events	Wrong final state or repeated cost	Use source version ID as idempotency key, immutable output keys and conditional DynamoDB writes.
Poison or malicious media	Retry storm/security exposure	Validate header/checksum, quarantine first, maxReceiveCount=5, DLQ alarm and controlled replay.
Sudden spend growth	Budget exceeded	CloudFront cache policy, lifecycle rules, concurrency limits, Budgets/Anomaly Detection and cost allocation tags.
Cold starts/quotas	Latency or throttling during spikes	Arm functions, small packages, reserved concurrency, quota review and measured provisioned concurrency only if needed.
Regional outage	Application unavailable	Accept in baseline; add S3 CRR, DynamoDB Global Tables and warm API stack when RTO/RPO is approved.
AI privacy or inaccurate labels	User trust or inappropriate metadata	Keep Rekognition optional, avoid face identification, apply confidence thresholds, review samples and support tag removal.
Vendor lock-in	Migration effort	Keep original media in S3-standard formats, export metadata, isolate AWS calls behind small service adapters and use IaC.

A. Suggested Delivery Sequence

Phase	Deliverable
1. Foundation	IaC, IAM, private S3, CloudFront/OAC, WAF, Cognito, logs and budgets.
2. API and upload	HTTP API, Lambda metadata service, DynamoDB, presigned upload and checksum validation.
3. Image processing	S3/EventBridge/SQS, image Lambda, DLQ, status model and failure tests.
4. Scale validation	10x load/burst tests, quota changes, cache tuning, cost-model calibration and operational runbook.
5. Extensions	MediaConvert video, approved Rekognition tagging, custom workers and multi-Region DR.

B. Operational Readiness and Governance

A serverless design reduces infrastructure administration, but it does not remove operational responsibility. Each delivery phase should therefore be implemented as version-controlled infrastructure as code and promoted through separate development, test and production environments. Before production release, the team should review the planned infrastructure change, verify IAM permissions, run API and event-contract tests, and confirm that dashboards, alarms and rollback instructions are present. This approach supports the operational-excellence guidance in the AWS Well-Architected Framework [1]. Application packages and transformation profiles should also be versioned so that a failed release can be rolled back without overwriting media generated by an earlier version.

Every alarm must lead to a named owner and a short runbook rather than only producing a notification. The first operational dashboard should show API latency and errors, Lambda throttles, SQS age of oldest message, DLQ depth, MediaConvert failures, Rekognition throttling, DynamoDB throttled requests and CloudFront cache-hit ratio. For example, an image-queue alarm should instruct the operator to check worker errors and concurrency, isolate a malicious or unsupported file, and redrive the DLQ only after the cause is fixed. Redrive remains safe because workers use the source object version and media ID as an idempotency key. This operational procedure complements the duplicate-event and partial batch-response behaviour documented for EventBridge and Lambda with SQS [3], [4].

Capacity governance is also required because managed services still have quotas. During Phase 4, the team should test at least the expected 10x burst, record the achieved Lambda concurrency and queue drain rate, and request quota increases before utilisation exceeds 70% of an approved limit. Reserved concurrency should protect the API Lambda from media workers, while the image, video and AI queues retain independent limits. Event schemas should include a version field and remain backward compatible during a rolling deployment. These controls prevent a rapid traffic increase or a new processor from creating an unplanned outage across the whole album.

C. Data Lifecycle, Privacy and Cost Guardrails

The metadata state machine should be treated as a business control: UPLOADING moves to QUEUED only after S3 confirms the object, PROCESSING identifies active work, and only an idempotent completion update can set READY or FAILED. Incomplete upload records can expire after a defined period, while originals follow the retention policy approved by the album owner. Derived thumbnails and video renditions can be regenerated, so they may use a shorter lifecycle than originals. A deletion request must remove CloudFront-accessible variants, the DynamoDB item and optional AI tags; versioned S3 objects then follow the approved permanent-deletion schedule. DynamoDB point-in-time recovery remains a recovery control, not a substitute for documented retention and deletion rules [9].

Rekognition remains an optional image-only path. The deployment should enable it only after the team agrees on permitted label categories, a confidence threshold, sample-based accuracy checks and a process for removing an incorrect tag. The proposed design performs general object labelling and does not introduce face identification. Access to originals is limited to the tagging role and approved prefixes, and the AI queue can be disabled without affecting resize or retrieval. These boundaries make the extension useful without treating AI output as automatically correct.

Costs are checked weekly against Table IV. Optional Transfer Acceleration, MediaConvert and Rekognition remain separately measurable so they can be limited when their benefit does not justify their unit cost [10]-[19].

D. Group Contribution Record

Member	Primary work	Shared responsibility
Le Thanh Hoang Minh - 106216845	Architecture and requirement traceability	Review all sections; present architecture and boundaries.
Quang Huy Tran - 106212636	Service rationale, security and reliability	Review all sections; present security/reliability.
Chau Chanh Nghi - 106216641	Sequence diagrams, cost and quantitative evaluation	Review all sections; present interactions and numbers.
Chung Hien Thong - 106212681	Alternatives, testing plan and final editing	Review all sections; present alternatives and trade-offs.

IX. CONCLUSION

The proposed design meets the new business requirements with a global edge, a serverless API and a decoupled media pipeline. Its most important decision is not an individual AWS product but the boundary between user interaction and media processing: direct S3 upload makes the API responsive, while EventBridge and separate SQS queues let image, video and AI work scale independently. DynamoDB fits the simple metadata model, and CloudFront addresses global read latency. Reliability and security controls are built into the flow rather than added later. The quantitative comparison favors serverless for this uncertain, rapidly growing workload, but the report also makes the regional-DR and cost risks explicit. The recommended next step is an IaC proof of concept followed by burst, failure, security and cost tests against the acceptance measures in Section VII-C.

REFERENCES

[1] Amazon Web Services, “AWS Well-Architected Framework,” Nov. 2024. [Online]. Available: https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html. Accessed: Jul. 28, 2026.

[2] Amazon Web Services, “Web application - Serverless Applications Lens.” [Online]. Available: https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/web-application.html. Accessed: Jul. 28, 2026.

[3] Amazon Web Services, “Amazon Simple Storage Service events - Amazon EventBridge.” [Online]. Available: https://docs.aws.amazon.com/eventbridge/latest/ref/events-ref-s3.html. Accessed: Jul. 28, 2026.

[4] Amazon Web Services, “Using Lambda with Amazon SQS.” [Online]. Available: https://docs.aws.amazon.com/lambda/latest/dg/with-sqs.html. Accessed: Jul. 28, 2026.

[5] Amazon Web Services, “Download and upload objects with presigned URLs.” [Online]. Available: https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html. Accessed: Jul. 28, 2026.

[6] Amazon Web Services, “Restrict access to an Amazon S3 origin.” [Online]. Available: https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html. Accessed: Jul. 28, 2026.

[7] Amazon Web Services, “Understanding Lambda function scaling.” [Online]. Available: https://docs.aws.amazon.com/lambda/latest/dg/lambda-concurrency.html. Accessed: Jul. 28, 2026.

[8] Amazon Web Services, “DynamoDB on-demand capacity mode.” [Online]. Available: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/on-demand-capacity-mode.html. Accessed: Jul. 28, 2026.

[9] Amazon Web Services, “Point-in-time backups for DynamoDB.” [Online]. Available: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Point-in-time-recovery.html. Accessed: Jul. 28, 2026.

[10] Amazon Web Services, “AWS Price List API,” regional product files for ap-southeast-2 and global services. [Online]. Available: https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/index.json. Accessed: Jul. 28, 2026.

[11] Amazon Web Services, “Amazon S3 pricing.” [Online]. Available: https://aws.amazon.com/s3/pricing/. Accessed: Jul. 28, 2026.

[12] Amazon Web Services, “AWS Lambda pricing.” [Online]. Available: https://aws.amazon.com/lambda/pricing/. Accessed: Jul. 28, 2026.

[13] Amazon Web Services, “Amazon API Gateway pricing.” [Online]. Available: https://aws.amazon.com/api-gateway/pricing/. Accessed: Jul. 28, 2026.

[14] Amazon Web Services, “Amazon CloudFront pricing.” [Online]. Available: https://aws.amazon.com/cloudfront/pricing/. Accessed: Jul. 28, 2026.

[15] Amazon Web Services, “Amazon DynamoDB pricing.” [Online]. Available: https://aws.amazon.com/dynamodb/pricing/. Accessed: Jul. 28, 2026.

[16] Amazon Web Services, “Amazon SQS pricing” and “Amazon EventBridge pricing.” [Online]. Available: https://aws.amazon.com/sqs/pricing/; https://aws.amazon.com/eventbridge/pricing/. Accessed: Jul. 28, 2026.

[17] Amazon Web Services, “AWS WAF pricing” and “Amazon Cognito pricing.” [Online]. Available: https://aws.amazon.com/waf/pricing/; https://aws.amazon.com/cognito/pricing/. Accessed: Jul. 28, 2026.

[18] Amazon Web Services, “AWS Elemental MediaConvert pricing.” [Online]. Available: https://aws.amazon.com/mediaconvert/pricing/. Accessed: Jul. 28, 2026.

[19] Amazon Web Services, “Amazon Rekognition pricing.” [Online]. Available: https://aws.amazon.com/rekognition/pricing/. Accessed: Jul. 28, 2026.

[20] Amazon Web Services, “AWS Architecture Icons.” [Online]. Available: https://aws.amazon.com/architecture/icons/. Accessed: Jul. 28, 2026.